ENCE260: Embedded Systems

Lecture 3: The UCFK4

Get to know your lab partner.

Revision

What is `(1<<5) | (1<<1)` ? Answer in binary, decimal, and hexadecimal.

$$\begin{array}{rcc} & 0010 & 0000 \\ | & 0000 & 0010 \\ \hline = & 0010 & 0010 \end{array}$$

= `0x22` Hexadecimal

32 + 2 = 34 Decimal

Revision

How would you configure and use (in code) an LED connected between `PB5` (supply) and `PD4` (ground)?

First, configure both as output, because it's an LED:

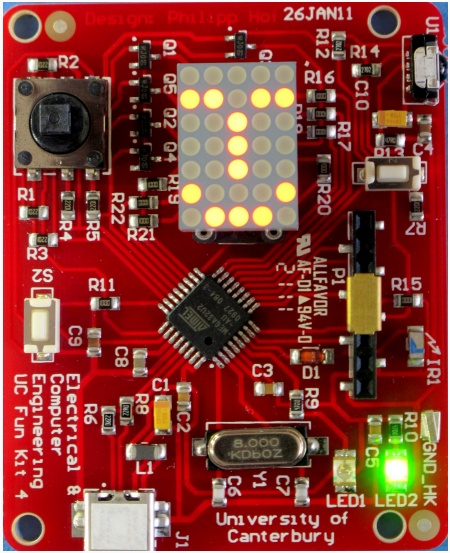
```
DDRB |= (1<<5);
DDRD |= (1<<4);
```

To use, we need to set them correctly (supply to high, ground to low).

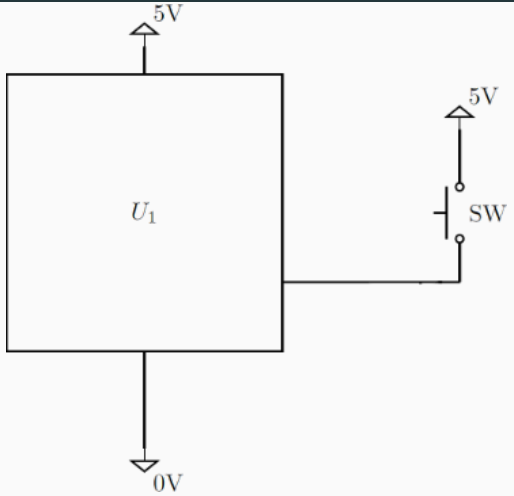
```
PORTB |= (1<<5);
PORTD &= ~(1<<4);
```

Reminders

- Labs start this week - make sure you have sourced a Fun Kit! Details on Learn.
- Whichever variant you have, make sure to bring a compatible *data* cable. *Test it.*
- Make sure you log into eng-git ASAP, so that you can be assigned a project repository.
- Assignment groups will be announced this week. Contact your partner ASAP! If they've not responded by the end of week 3, let me know by email.

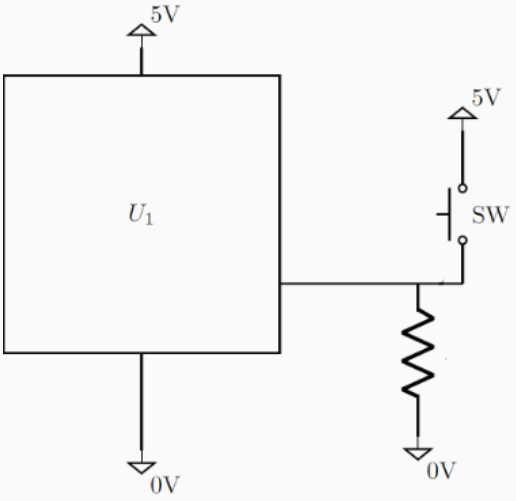


Switch Default Values



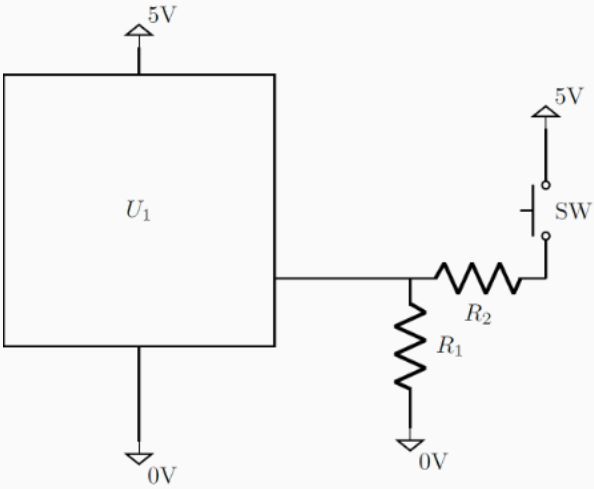
- When switch closed, this pin is at $5\,\mathrm{V}$.
- When switch open, this pin is at ??? V.
- Unconnected wires can be at indeterminate voltages - they act like antenna.
- If active low (connected to ground when closed), the switch would never work.

Pull-Up and Pull-Down Resistors



- We can give a pin a *default* value by connecting a resistor between it and supply/ground.
- This works because the current through an input pin should be very small.
- The default value should be the opposite of the active value.
- Pull-up resistors for active-low switches, pull-down resistors for active high switches.

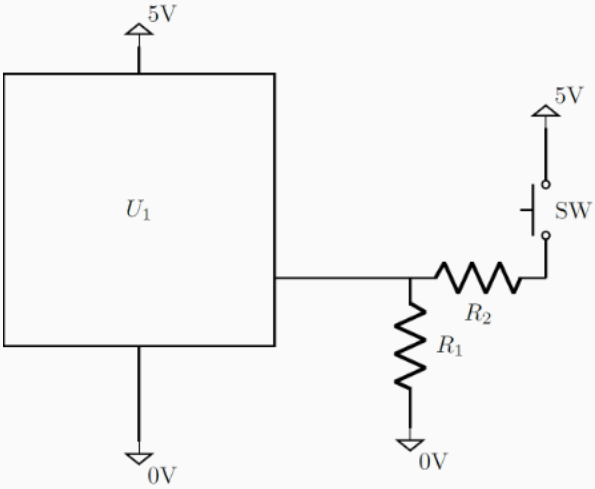
Current Limiting



- If an input pin is misconfigured as an output, a short can occur.
- To stop the microcontroller from blowing, it can be a good idea to install a current-limiting resistor on switch pins as well as LED pins.
- This can be on either side of the pull-up/down.
- These are also called guard resistors.

Active High Switch

To check the `PD7` switch on the UCFK4 we need to:



- Configure `PD7` as an input.

```
DDRD &= ~(1<<7);
```

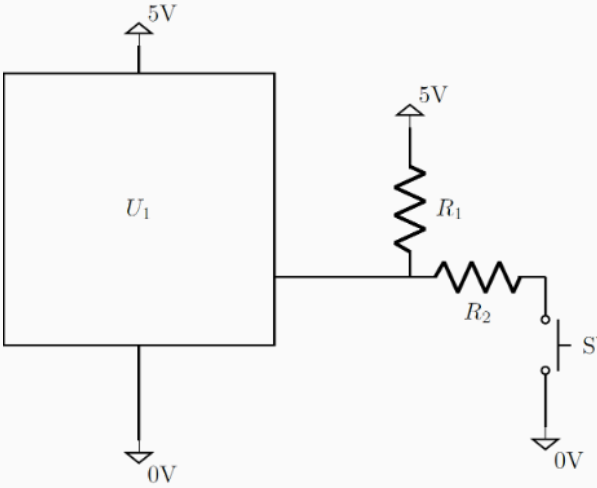
- Test state of `PD7`.

```
if ((PIND & (1<<7)) != 0) {
```

- Leave the other bits in any modified registers the same.

Active Low Switch

To check an active low switch on `PD7` we need to:



- Configure `PD7` as an input.

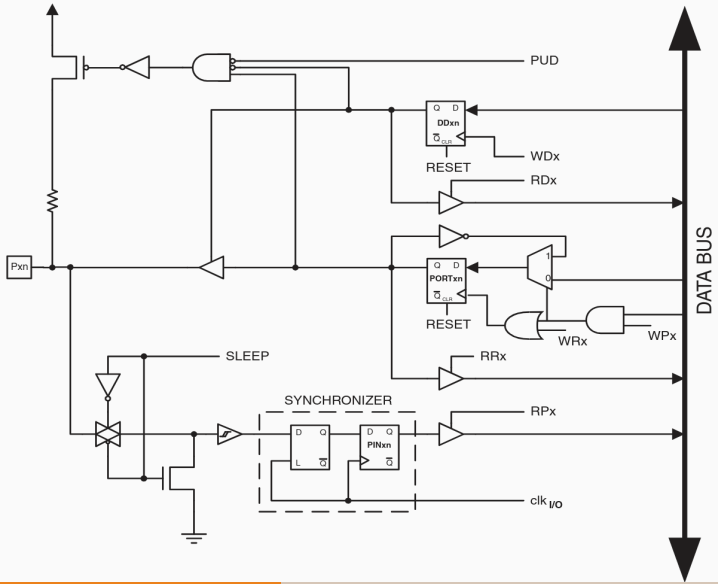
```
DDRD &= ~(1<<7);
```

- Test state of `PD7`.

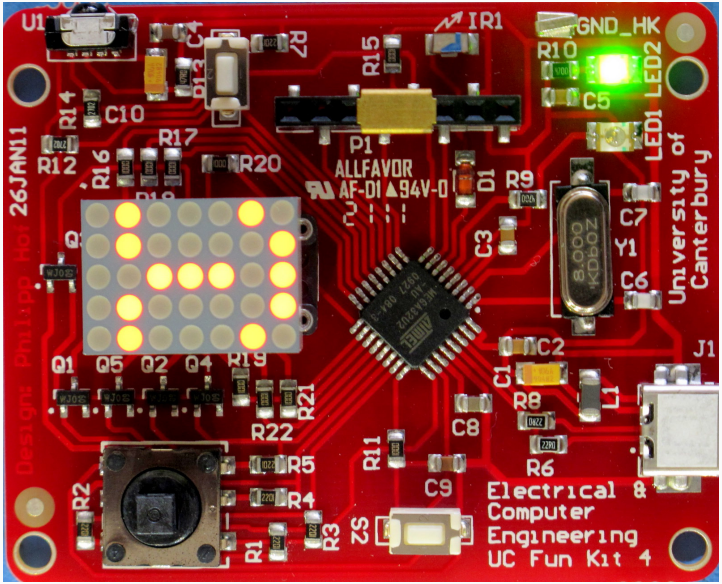
```
if ((PIND & (1<<7)) == 0) {
```

Leave the other bits in any modified registers the same.

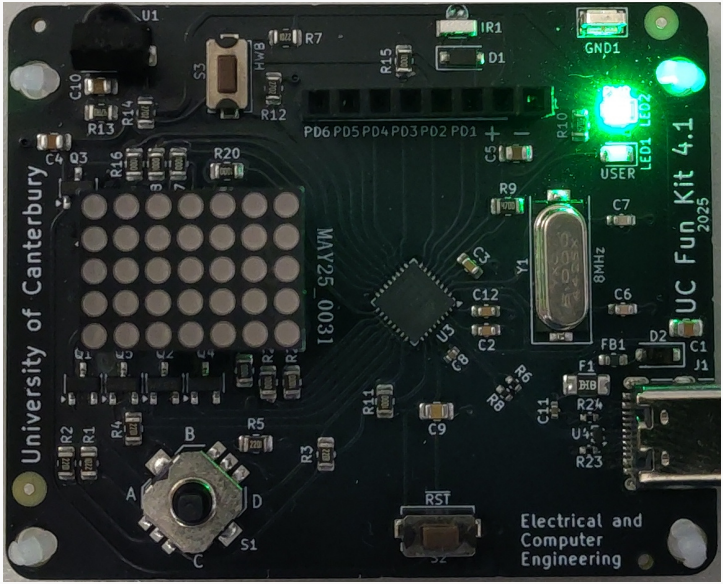
Internal Pull-Up Resistors



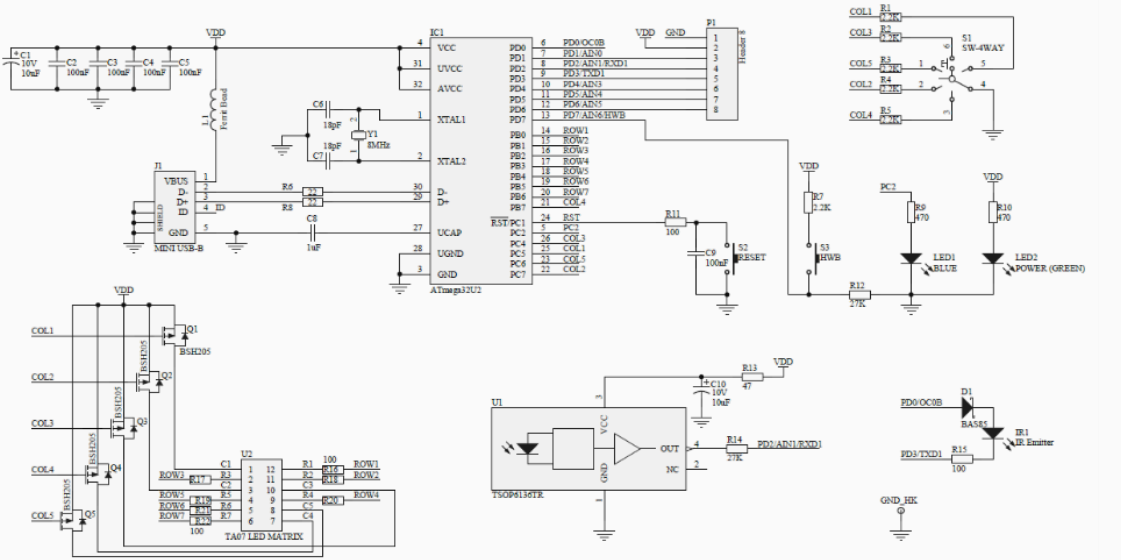
UCFK4


U1
IR1
GND_HK
R10
LED1
LED2
C5
C4
R7
R15
R14
C10
R12
R17
R16
R20
P1
ALLFAVOR
AF-01▲94V-0
D1
R9
C7
Y1
C6
C3
University of Canterbury
26JAN11
Designed Phillip Moi
Q1
Q5
Q2
Q4
R19
R21
R22
C2
J1
C1
R8
L1
C8
R6
R11
R5
R4
R2
R3
R1
C9
ZS
Electrical & Computer Engineering
UC Fun Kit 4

UCFK4.1


University of Canterbury
UC Fun Kit 4.1
2025
MAY25_0031
Electrical and
Computer
Engineering
PD6 PD5 PD4 PD3 PD2 PD1
GND1
USER
LED1
LED2
IR1
D1
RST
S1
S2
S3
U1
U3
U4
Y1
8MHz
J1
A
B
C
D

Schematic


VDD
C1 10V 10nF
C2 100nF
C3 100nF
C4 100nF
C5 100nF
L1 Ferrit Bead
IC1
VCC
UVCC
AVCC
XTAL1
XTAL2
UCAP
UGND
GND
PD0/OC0B
PD1/AIN0
PD2/AIN1/RXD1
PD3/TXD1
PD4/AIN3
PD5/AIN4
PD6/AIN5
PD7/AIN6/HWB
PB0 ROW1
PB1 ROW2
PB2 ROW3
PB3 ROW4
PB4 ROW5
PB5 ROW6
PB6 ROW7
PB7 COL4
RST PC1 PC2
PC4 COL3
PC5 COL1
PC6 COL5
PC7 COL2
ATmega32U2
C6 18pF
C7 18pF
Y1 8MHz
J1
VBUS
D-
D+
ID
GND
SHIELD
MINI USB-B
R6 22
R8 22
C8 1uF
P1
Header 8
R11 100
C9 100nF
S2 RESET
R7 2.2K
S3 HWB
R12 27K
R9 470
R10 470
LED1 BLUE
LED2 POWER (GREEN)
COL1
COL2
COL3
COL4
COL5
R1 2.2K
R2 2.2K
R3 2.2K
R4 2.2K
R5 2.2K
S1 5W-4WAY
Q1 BSH205
Q2 BSH205
Q3 BSH205
Q4 BSH205
Q5 BSH205
U2
TA07 LED MATRIX
ROW1
ROW2
ROW4
ROW5
ROW6
ROW7
R13 47
C10 10V 10nF
U1
TSOP5136TR
VCC
GND
OUT
NC
R14 27K
PD2/AIN1/RXD1
D1 BAS85
IR1 IR Emitter
PD3/TXD1
R15 100
GND_HK

Component List

- **Crystal:** 8 MHz, sets the clock frequency.
- **Reset Button (S2):** Resets the microcontroller.
- **Pushbutton (S3):** `PD7` is high (1) when button is pressed.
- **NavSwitch (S1):** Can be pushed in five directions (North, East, South, West, Down). Shares pins with the LED matrix.
- **USB Connector:** Provides 5 V power to the board and communication with the AVR DFU bootloader.

Component List (Cont.)

- **Green LED:** Pilot light; always lit when board is powered.
- **User LED:** Driven by setting `PC2` as a high output. Blue for V4, white for V4.1.
- **LED matrix:** 7 rows $\times$ 5 columns. An individual LED is turned on by driving both the corresponding row and column low. Colours vary between boards.
- **Infrared LED:** For sending data. Driven by setting `PD0` high output and `PD3` low output.
- **Infrared receiver:** For receiving data. Connected to `PD2`.
- **Expansion connector (P1):** Allows connection of external devices. e.g. a buzzer for sound.

UCFK Software

- All Erskine lab computers have AVR tools installed (`avr-gcc`, `avr-libc`, `dfu-programmer`, etc.).
- Learn has instructions for installing these your own machine (Linux, Windows, Mac).
  - Note we don't provide support to help you install software on your own machines. We provide the Erskine computers which have the software installed.
  - The embedded systems assignment will be assessed on the Erskine computers, which run `avr-gcc` 7.3.0. Different versions may generate additional errors - check your code on the correct version.
- The UCFK repo, which contains labs, device drivers, apps etc. can be cloned from git with the following command. You have to log into eng-git first!

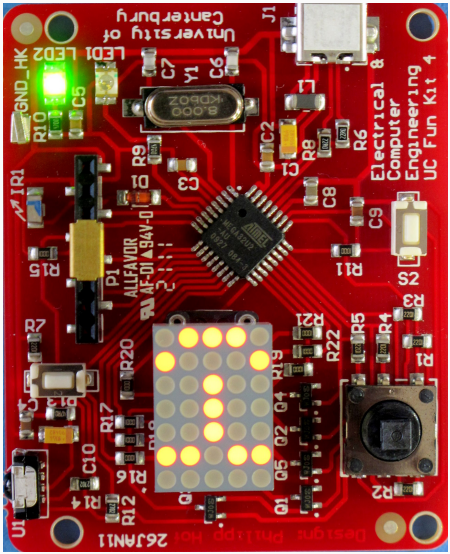
```
git clone https://eng-git.canterbury.ac.nz/ssp63/ence260-ucfk4.git
```

`make`

- `make` is a command-line program that “automatically” compiles your program for you.
- It does this by reading rules from a `Makefile`, and running the commands that need to be run.
- Commands beyond compilation can also be added. For example, in all the supplied apps and lab programs, `make program` will compile an executable *and* program the resultant `.hex` onto the micro.
- Note that in order to program an application you will need to first reset the UCFK.

Resetting and Restarting

- The reset button is the one nearest the USB.
- You can restart the micro, which will run the program currently in flash, by:
  - Hold the user button (S3).
  - Tap the reset button (S2).
  - Release S3.
- Note that the restart actually happens once S2 is released. Therefore, don't use S3 near the start of a program.



Lab Programming Tips

- Don't move or rearrange files within the given directory structure; there are dependencies that cannot be easily changed. All labs and apps should compile where they are.
- Don't have multiple instances of the same file open at once (e.g., `led.c` & `button.c` exist in both `lab1-ex3` and `lab1-ex4`). It's easy to get stuck by editing a file from a previous question.
- Read the instructions more than once if needed, and have the lectures for reference.

Programmer Warnings

- After loading your program onto the Fun Kit, you may see the message `dfu-programmer failed to release interface 0`
- This is caused by the micro resetting itself once programming has finished, which means the USB controller can't safely remove the connection in the usual way.
- You can safely ignore this error throughout this course - it can only happen if the programming succeeded.

Assignment Details

- Your assignment is to develop a **2-player**, **2-board** game for the UCFK.
- $50\,\%$ of the marks will be for the demonstration, $50\,\%$ for code quality.
- Your game should show high use of communication between the two boards, since this best demonstrates management of hardware, scheduling robustness, and partner work.
- If you are repeating the course, you have to redo the assignment and cannot use your code from the previous year.
- The assignment specification and marking rubric are on Learn.

Key Concepts

- Input pins can be given default values with pull-up and pull-down resistors.
- Besides switches, the Fun Kit has a number of key components, which you need to be able to identify.
- Programs for the Fun Kit are in specific relative locations within the repository, and can be compiled/programmed with `make`.

Homework

Compare the appropriate schematic on Learn with your Fun Kit, and make sure you can point out all of the major (listed) components on each.

Also, make sure you finish all of lab 1 this week - lab 2 tends to take a fair bit longer.